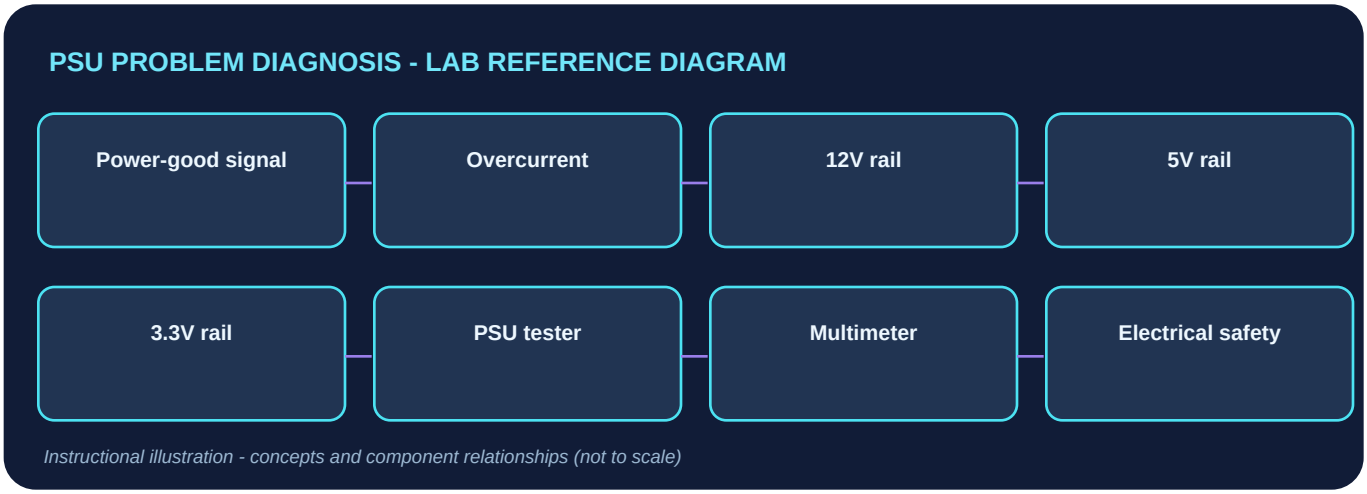
3.1.9 Troubleshoot Power Supply Problems

STUDENT STUDY GUIDE + VOCABULARY + VISUAL REFERENCE

Learning goal

Test and isolate PSU faults.

Reference image



Use the labeled illustration to identify the key components, commands, or checkpoints you will encounter in this lab.

Prepare for the simulation

1. Gather power-related symptoms and recent changes.
2. Check cables and correct PSU wattage.
3. Use an appropriate tester without opening the PSU.
4. Verify the repair under a controlled load.

Troubleshooting and safe practice

• Shutdown under load: consider PSU overload or overheating.

• Electrical safety: never probe inside a PSU case.

3.1.9 | VOCABULARY & REVIEW

Essential vocabulary

Term	Meaning / recognition cue
Power-good signal	Electrical power characteristic or component; assess safely with rated tools and correct connectors.
Overcurrent	A key term in psu problem diagnosis; identify its purpose, where it is used, and how to verify it.
12V rail	Electrical power characteristic or component; assess safely with rated tools and correct connectors.
5V rail	Electrical power characteristic or component; assess safely with rated tools and correct connectors.
3.3V rail	Electrical power characteristic or component; assess safely with rated tools and correct connectors.
PSU tester	Observation or tool used to identify faults and confirm the repair.
Multimeter	A key term in psu problem diagnosis; identify its purpose, where it is used, and how to verify it.
Electrical safety	A key term in psu problem diagnosis; identify its purpose, where it is used, and how to verify it.

Knowledge check - answer in your own words

1. Identify two components or commands in the reference diagram and explain what each does.

2. What should you verify before changing or installing hardware, settings, or files in this lab?

3. Choose one listed failure symptom. What is your first safe diagnostic step?

4. How will you prove the task is complete, rather than merely assuming success?

Lab readiness checklist

[ ] I can identify the labeled reference items. [ ] I understand the safety precautions.

[ ] I can explain the vocabulary. [ ] I can verify and document successful results.

Student: _____ Date: _____